

SARVOTTAM MATHEMATICAL NOTATION BENCHMARK TEST

1. Exponents: Evaluate $x^2 + y^3 = z^5$ when $x = 10 \times 10^6 \text{ M}$ and $f(x) = e^{(\tan^{-1} x)}$.
2. Subscripts: Reaction of $2\text{H}_2\text{O} + \text{AlCl}_3$ with base at constant K_b .
3. Fractions: If vernier constant is $(1 / 10\text{N})$ and ratio is (a / b) .
4. Square Roots: Find the value of $\sqrt{x} + 12\sqrt{3}$ when $x = 27$.
5. Greek Letters: Parameters $\alpha = 0.05$, $\beta = 1.2$, $\theta = 45^\circ$, $\lambda = 600 \text{ nm}$, $\pi = 3.14159$, and $\Delta E = h\nu$.
6. Inequalities: Condition $x \leq 10$, $y \geq 0$, tolerance ± 0.01 , with integral $\int f(x) dx$.